File Edit Selection View Go Run Terminal Help

HW

EXPLORER

- HW
  - HW2
  - HW3
  - HW5
    - WilliamsBenjamin_HW5
      - Exercise1
        - exercise1.py
        - vehicles.csv
        - vehicles.names
      - Exercise2
      - Exercise3
    - exercise3.py
    - vehicles.csv
  - WilliamsBenjamin_HW1

exercise1.py

HW5 > WilliamsBenjamin_HW5 > Exercise1 > exercise1.py > ...

```python
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

df = pd.read_csv("vehicles.csv")
if "make" in df.columns:
    df = df.drop(columns=["make"])

y = df["mpg"].values
X = df.drop(columns=["mpg"])

X = pd.get_dummies(X, drop_first=True)

model = LinearRegression()
model.fit(X, y)

coefs = pd.Series(model.coef_, index=X.columns)
top5 = coefs.abs().sort_values(ascending=False).head(5)

print("Top 5 most important (by absolute weight):")
for name, val in top5.items():
    print(name, coefs[name])
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW$ cd HW5
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW5/WilliamsBenjamin_HW5/
Exercise1/exercise1.py
Top 5 most important (by absolute weight):
wt -3.7153039283274794
am 2.5202268872084304
qsec 0.8210407496746295
drat 0.7871109722361131
gear 0.6554130170817877
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5$
```

bash Machi...
Python HW5

Do you want to install the recommended 'Rainbow CSV' extension from mechatroner for vehicles.csv?

Install Show Recommendations

OUTLINE
TIMELINE

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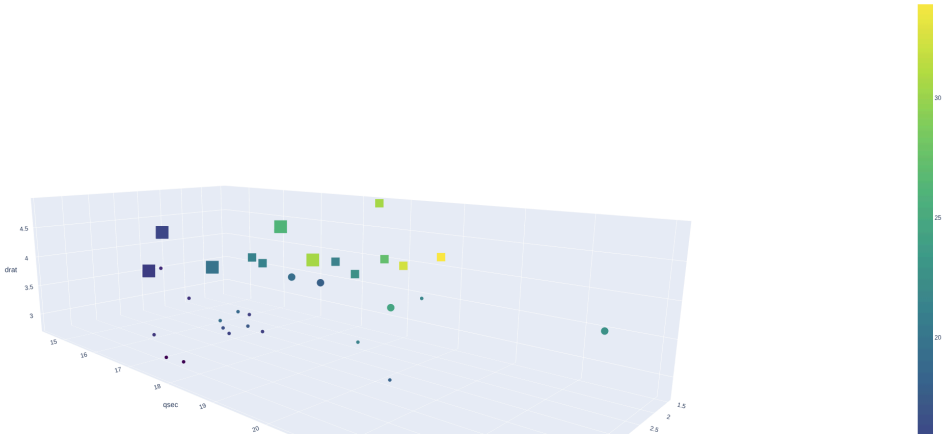
Feb 25 22:03

HW5_Exercise2_6DPlot.

File /home/ben/Documents/Development/MachineLearning/HW/HW5/HW5_Exercise2_6DPlot.html

6D Plot (x=wt, y=qsec, z=drat, color=mpg, size=gear, shape=am)



File Edit Selection View Go Run Terminal Help

HW

EXPLORER

- HW
  - HW2
  - HW3
  - HW5
    - WilliamsBenjamin_HW5
      - Exercise1
        - exercise1.py
        - image.png
        - vehicles.csv
        - vehicles.names
      - Exercise2
        - exercise2.py
        - vehicles.csv
        - vehicles.names
      - Exercise3
    - exercise3.py
    - HW5_Exercise2_6DPlot.html
    - vehicles.csv
  - WilliamsBenjamin_HW1

exercise1.py | image.png | exercise2.py | vehicles.names

HW5 > WilliamsBenjamin_HW5 > Exercise2 > exercise2.py > ...

```python
import numpy as np
import pandas as pd
import plotly.graph_objects as go

df = pd.read_csv("vehicles.csv")

req = ["mpg", "wt", "am", "qsec", "drat", "gear"]
missing = [c for c in req if c not in df.columns]
if missing:
    raise ValueError(f"Missing columns in vehicles.csv: {missing}")

x_name = "wt"
y_name = "qsec"
z_name = "drat"
c_name = "mpg"
s_name = "gear"
shape_name = "am"

x = df[x_name].to_numpy(dtype=float)
y2 = df[y_name].to_numpy(dtype=float)
z = df[z_name].to_numpy(dtype=float)
color = df[c_name].to_numpy(dtype=float)

sz = df[s_name].to_numpy(dtype=float)
mn, mx = float(np.nanmin(sz)), float(np.nanmax(sz))
ms = 8 + 18 * (sz - mn) / (mx - mn + 1e-12)

am_vals = df[shape_name].to_numpy()
symbols = np.where(am_vals == 0, "circle", "square")

fig = go.Figure(
    data=[
        go.Scatter3d(
            x=x,
            y=y2,
            z=z,
            mode="markers",
            marker=dict(
                size=ms,
                color=color,
                symbol=symbols,
                colorscale="Viridis",
                showscale=True,
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW5/WilliamsBenjamin_HW5/
Exercise2/exercise2.py
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW5/WilliamsBenjamin_HW5/
Exercise2/exercise2.py
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW5/WilliamsBenjamin_HW5/
Exercise2/exercise2.py
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW5/WilliamsBenjamin_HW5/
Exercise1/exercise1.py
Top 5 most important (by absolute weight):
wt -3.715303928327479
am 2.5202368872084364
qsec 0.8214847496746295
drat 0.7071997223611131
gear 0.6554130778017877
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW5/WilliamsBenjamin_HW
Exercise2/exercise2.py
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5$
```

Do you want to install the recommended 'Rainbow CSV' extension from mechatroner for vehicles.csv?

Install Show Recommendations

Ln 27, Col 1 Spaces: 4 UTF-8 LF {} Python Python 3.14.3

File Edit Selection View Go Run Terminal Help

HW

EXPLORER

- HW
  - HW2
  - HW3
  - HW5
    - WilliamsBenjamin_HW5
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        - exercise1.py
        - image.png
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        - exercise3.py
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    - HW5_Exercise2_6DPlot.html
    - vehicles.csv
  - WilliamsBenjamin_HW1

exercise1.py | image.png .../Exercise1 | exercise2.py | image.png .../Exercise2 | exercise3.py

HW5 > WilliamsBenjamin_HW5 > Exercise3 > exercise3.py > ...

```python
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression

df = pd.read_csv("vehicles.csv")
if "make" in df.columns:
    df = df.drop(columns=["make"])

y = df["mpg"].values
X_raw = df.drop(columns=["mpg"])
X = pd.get_dummies(X_raw, drop_first=True)

model = LinearRegression()
model.fit(X, y)

coefs = pd.Series(model.coef_, index=X.columns)
top5_names = coefs.abs().sort_values(ascending=False).head(5).index.tolist()

x_new_full = np.array([[6, 163, 111, 3.9, 2.77, 16.45, 0, 1, 4, 4]], dtype=float)

x_new = pd.DataFrame(columns=X.columns)
x_new.loc[0] = 0.0

vals = dict(zip(df.drop(columns=["mpg"]).columns, x_new_full.flatten()))
for k, v in vals.items():
    if k in X_raw.columns:
        if k in x_new.columns:
            x_new.loc[0, k] = float(v)

x_new_top5 = x_new[top5_names].values
pred = model.predict(x_new)[0]

print("Predicted mpg:", pred)
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
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ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5$ ^C
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5$ ^C
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/HW/HW5/WilliamsBenjamin_HW
Exercise3/exercise3.py
Predicted mpg: 22.0525233652674
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning/HW/HW5$
```

bash Machi...
Python HW5

Do you want to install the recomm
extension from mechatroner for v

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